

ActInf Textbook Group ~ Cohort 7 ~ Session 24

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All right. Hello everyone. Welcome back to the active inference uh institutees textbook reading group. Uh today this is cohort 7 wrapping up with chapter 10. It's been quite a journey through uh the first part of the book which as we know is a bit more focused on formalisms on theory on anything from marov blankets to wrapping our heads around hierarchical uh variational message passing in microcortical circuits as well as looking at sort of a basian framework for doing updating and then linking that to action and perception and action perception loops. uh from the active inference framework perspective. And then with the second part of the textbook, we saw a lot more in terms of application models. We had a recipe for active inference ranging from uh defining our generative models, our generative processes, considering different kinds of components for models, likelihood and state transition matrices, looking at the discrete case versus the continuous case and all the rest. And then our last chapter for this cohort prior to chapter 10, chapter nine, we were looking at what what happens whenever we want to actually fit models to empirical data. Whenever we want to do group analysis, uh, and and kind of wrapping our heads around the idea of composing a subjective model for our agents, then having our own kind of model of the agents model. So something like metabasian modeling or analysis. Now the chapter 10 which kind of sticks out a little bit uh a bit of an outlier in the the second part of the book. It's almost as if they could have written a third part of the book that would have been labeled something like you know active inference and its connections. We saw in this chapter that we're actually no longer directly talking about kind of the mechanical or or sort of of of uh engineering aspects of creating or testing or fitting models. Now we're just kind of wrapping up with looking at the inter relationships between active inference and a variety of other fields. Both fields from which active inference kind of grew as well as fields to which it can contribute and uh and those which fall within both of those prior categories. So in uh the introduction to chapter 10 we're uh described uh it's described to us active inference is a complete solution to sentient

organisms adaptive problems which attempts to provide a unified perspective on action selection perception attention and even emotion regulation. Uh that list could continue on and on actually for those of us who have kept up throughout the book. Um and it's kind of reiterating you know we know that active inference is essentially a unified it's a normative framework operating from first principles brings back the old uh neats and scruffies uh argument or debate from the first chapter of the book where we are in the realm of the neats although we certainly would not uh deny a scruffy or two uh especially to welcome them into the fold if they so chose um in In section 10.3 there's a pretty inf interesting quote uh from Daniel Dennett the late Daniel Dennett of course and uh his suggestion to kind of model the whole iguana uh with active inference uh we we we do something like that in an attempt to create proxies of reality whether it be creating a model to simulate in silicone or it be to fit a model to a a real world organism or phenomena or otherwise where here free energy minimization acts as a kind of first a first principle by which we can derive implications about specific cognitive processes in their neuronal underpinnings. So active inference takes a kind of inactive integrative approach to things like preference realization, self-evidencing, self-preservation. And we're reminded of homeostasis or or as well as allostasis or a kind of homeostasis of the future. Uh the idea that you know we have bound there's bounded rationality that that organisms and agents and so on kind of exist within their immediate context. They come with you know their their own histories or memories and and beliefs that they're you know actively kind of updating and and also testing through action. And then um we we're we're reminded again of the self-evidencing aspect of action perception. Both of these are self-evidencing in the sense that they both lead to the agent aligning its expectations given its generative model by either changing the world or changing its mind. So memory and attention also are thought to be optimizing the same objective. Long-term memory develops through learning model parameters. Remember that learning operates at a somewhat slower time scale than than inference itself which inference is usually seen as kind of a real time moment to moment process. Whereas learning could happen it was just slowly or you could imagine like it's almost like a a mouse in the T-maze uh moment by moment position from position uh or to position in the T-maze as they move left or right or visit the informative queue at at the speed of real-time inference. You might say, you know, perhaps after the entire trial, you know, they achieved the cheese or unfortunately they didn't and they, you know, obtained a a negative stimulus. Uh, in either case, it's kind of like maybe after the whole trial, that's where you kind of take in the whole experience as it were and maybe that's where

learning happens or maybe it happens at another time or maybe it happens during sleep, you know, which which we certainly uh interestingly do. Of course, our brains are never fully off, right? So there there's variety of processes that occur during sleep and including uh the kind of modeling terms would be both parameter learning as well as structure learning. You know it's interesting to see what a night of sleep does you whenever you uh are are working on a problem and then suddenly the next day maybe you feel as if you've realized the answer or at least have found some kind of clarity that you didn't have prior. Um see we're also reminded of uh you know the the deep relationship between active inference and when we're talking about action perception planning all these kinds of things they not just look at you know neuronal processes in the brain but it also has heavy overlap with a lot of uh ideas from of course cognitive science as well as kind of uh psychology in a broader sense. You know there there are interesting papers for those who might be interested in uh psychology and you know there's uh um some interesting work that's been done on trying to relate you know that active inference though it is operating first principles it of course I mean the whole point of chapter 10 is not to say oh this is a first principles kind of framework and therefore we reject all else instead it is it is kind of opening doors to interdisciplinary um uh communication, dialogue and so on. And so it's just you know interesting work that's been done on trying to make the you know make the point that there sometimes it's more of a matter of ontology your your terminology you use and your respective onlogical take or or what words are in your vocabulary for describing different kinds of processes or phenomena or perhaps even the word process has a different meaning from one field to another, right? Um but uh nonetheless they do uh kind of discuss some other things that that I always kind of emphasize whenever I describe active inference to others. You bec because with with the kinds of work that can be done in active inference, say in psychology or something like precision psychiatry and you know the the notion that that a neurotransmitter might be modeled as a kind of precision or attention related or or synaptic gain related parameter over a distribution in a model. Um, you know, we're we're we're we're finding, especially today, the relationship between a lot of these kind of cognitive science and and and kind of psychological sciences like concepts and notions, but actually being able to find different ways to approximate them or or or even test them and fit them to experimental data using modern machine learning, reinforcement learning. Active inference in some ways can be interpreted as a kind of variation of reinforcement learning. Um, and so it's it's just kind of leveraging modern tools, technologies, and so on, not just to try and create the next kind of superhuman AI, but to actually find ways

where possible to kind of, you know, model the dynamic or or or causal relationships um in in living organisms or or uh in broader processes or modeling, you know, potentially entire uh societies in some way and looking at collective emergent collective behavior otherwise and so it's I like to emphasize sort of um the work of someone who collaborated with uh Carl Friston at one point in time uh uh Yori Busousaki uh who who's a I took a lot of influence from he's done a lot of work in kind of like rhythmic uh processes in in in in neuronal activity looking at things like gamma versus beta and alpha uh frequency bands as has kind of extracted through uh and and engaged in with like spectral analysis when looking at neuroimaging data. And uh he and many others have been making the point that you know the cognitive and psychological sciences should perhaps today be a little bit more wary of uh of you know a really brilliant mind that was William James in in the 19th century. But but we should maybe be wary of kind of a lot of these inherited Jamesian categories which quote despite their great historic value may be quite arbitrary or they may not correspond to separate cognitive and neural processes. So the authors here again they they reference Busaki as well for example whose work suggests a more critical stance towards using James as a theoretical springboard from which to kind of confine our empirical undertakings. Though James' work should not be neglected per se, Busaki advised an inside out approach to neuroscience and neurobiology such that mechanisms and neural underpinnings are studied at or alongside if not prior to determining the supposed cognitive mechanisms mechanisms in play. And you know there's a kind of um you know in an interview he said like oh we'll take any you know college level psychology class and look at a lot of the terminology that's being used then and then go to you know some of William James's classic work and have a look at the table of contents and you'll realize there's a lot of the same terms that we're using today right like perception and perception kind of you know is broken into so many different sorts of meanings and and uh is used differently and there you know debates over what perception is and active inference kind of inserts itself into that debate to say, "Oh, yeah, there's perception, but we don't just passively perceive things, which is kind of a a kind of uh artifact of the the 19th century and and its its take on on something like what was emerging in it later called psychology, which is, you know, the idea that we're just kind of passively taking in sensory stimuli. Uh, instead we're acting in the world, right? and we pick up objects and we look at them from different angles and we sort of test our beliefs, you know, and that's part of the learning or and inferential process. And again, that kind of bounded rationality notion, right? We're in bodies and we're, you know, we're functioning. Um, all this is to say, uh, it it it's it's interesting one to to kind of kind of situate

active inference within a broader historical context, one that is contending with sort of, you know, the history of of of of psychology and the in and the social and behavioral uh, and cognitive sciences and also looking at it in the context of, you know, today the world of of AI where you could you could effectively make an active inference agent who who does a variety of things that anyone would want them to do as far as something like reinforcement learning goes, right? Like they're they're taking in data or observations and then they're, you know, doing different kinds of inferial processes and then emitting some kind of output depending on what kind of actuators you connect to it, right? So if you get it, you give it arms and then it can maybe move its arms um and and so on. So, so the the book nonetheless is is trying to kind of kind of rally together a lot of these different disciplines, you know. So, we we look at um anything from cybernetics and kind of control theory. We look at its relation to more general notions of of predictive processing uh which is a rather broad category also relationship to predictive coding is a sort of subset or variation of predictive processing. uh we're you know kind of re- reintroduced to the basian brain hypothesis which is you know heavily related of course the notion of the brain uh does something like you know it's it's it's it's still a matter of debate if if we're saying it expressly does this or if it does something like but we could say at minimum for active inference it does something like updating its beliefs using some kind of prior information right in order to infer some sort of updated belief or a posterior uh belief And then uh we're looking at action control theory, idiom motor theory reminded of you know um proprioception and and kind of you know motor commands and movement uh utility and decision making which then can start to kind of actually look like something like economics and we know that uh there are these risk terms that are kind of interpreted from some variation the uh variations of the expected free energy formula. Um again behavior bounded rationality um veilance emotion and motivation. Uh I've personally always kind of found that interesting and that touches kind of on psychology again uh and and cognitive science. But uh I I've I've repeatedly shared this paper with others just because I think it's kind of enlightening for those who are kind of starting to get more deeply into active inference. It seems like people commonly come across this one uh deeply felt affect which is a few years old now. Um but it it just you know a lot of these kind of names on the paper are you know some of the kind of the early kind of who have done a lot of the the kind of heavy lifting with developing active inference and and sort of finding ways to formalize it, advance it, uh teach it and uh who are still very active today. And this is a nice like you know in some ways it falls in line with with in a effective neuroscience the work of uh there are some who would not like that I I say this

but I still see the connection with uh with something like socially constructed affect or not affect but rather emotion whenever trying to talk about like what is emotion what is you know what what does that mean how does it work so we have you know in some variations we have like um punk concept and the idea that emotion can be broken into like you know seven or so discrete categories and then any kind of emotional you know expression that you see someone have or they say they're feeling some kind of emotion it's going to be somewhere in the realm of those seven categories as if they were like clusters that we're extracting from a bunch of noisy data or something like that. um you know that that would then be linked kind of to the more archaic area of our of our brains and be related to evolution. And we have this and then there are others who you know do this more socially constructed idea of emotion where actually it aligns with active inference as far as my take on it goes which is that the idea is that actually it's your body and interoception that generate a significant amount of observable signals that we can then observe. So it's actually it's not that we start with feeling sad and then we feel a tightness in the stomach or you know some other kind of bodily feeling you know a feeling of exhaustion or instead it's the other way around it's like we start feeling the observations first then we start to infer like oh I must be sad right like that it's actually that kind of direction and it's quite similar to active inference whenever we think of state in uh state inference right because the idea is that you're taking in observations and then you infer a kind of hidden state of the world or belief including beliefs about one's own body and then of course we have you know human language is you know a way or attempt to articulate what those feelings are and so actually it's saying you're sad is coming after all of that right um is one take in any case um you know it's it's very interesting to see like this is one example of kind of how using an active inference sort of framework can help us to interpret or reinterpret a broad variety of phenomena just by kind of relying on or sticking to some of these normative framework like you know perspectives on what is what is hidden state inference and what does that say about about uh emotion or emotionality and then um how do we how do we construct those things and how do we you know how do we model them and so I just thought I'd spend an extra minute on this just because it's sort of an example example of you know the many active inference papers that have been published. It's like well yes we can talk about these sort of everyday phenomena seemingly everyday phenomena uh you know like like emotion or or or affect but then what we can do is actually start to sort of formalize different kinds of models consider like okay well if we did have to make a model that was simple as possible but no simpler um what what does that look like in order to

properly model a motion beyond just you know at some point the ideal is to fit it to empirical data but even getting you know Even prior to that, how do we how do we even start imagining this? We've had questions in the past from learners who say, you know, what what if you have bad what if you're modeling something? What if your priors are just bad in the sense of like what if what if they don't work appropriately or what if they don't lead to the sort of the optimal or expected outcome that you know that we have? Um, you know, isn't that kind of a funny thing to, you know, include poor priors in the beginning? And yet, I mean, whenever we're getting started on modeling, we just don't we don't know, right? You you use prior literature that's been published or or developed and ideally, you know, peer-reviewed and and you move in the direction of, you know, you have a hypothesis and you so you end up having your own hypothesis about the model. uh and then and then you you kind of try different things and then you try to fit it to empirical data. Uh and and that's sort of the interesting aspect of of active inference and and other models that kind of work similarly is that you know it's you might start with bad priors but over time uh if if the if the model engages in different kinds of inference and learning you know processes then ideally it gets better. In the same way that someone who doesn't know calculus can can sit down and start studying calculus. They have awful priors presumably because they haven't learned anything about it yet. And over time they learn it, right? Hopefully. So that's kind of the, you know, we find ourselves on the outside here. we are constructing our own kind of objective model or almost think of it like we're creating an object out of you know an agent and and and trying to understand them and kind of how they work on the inside right um so uh anything else um you know I think I've largely covered a lot of this we also talked about um attention salience and epistemic dynamics which is an interesting piece you know and we didn't cover these terms too strongly in this in this uh particular run of the textbook. But uh we know that that attention is related to to something like precision. So the idea of like kind of sharpening one's precision in the sense of uh you know how how certain are you uh about your your beliefs or how certain are you about about the the the stimuli that you're receiving, what you're observing, right? You imagine someone who's kind of sleepy and so they're not fully taking in what's going on uh in their in their visual field for is a kind of a crude quick example, right? Versus someone who's, you know, fully taking in uh sort of the moment, but these very kind of, you know, everyday use of terms here. Um but but we've already we've already made it through a lot of the kind of mechanical uh aspects and and formula and hurdles in the textbook up to this point. So I'm just avoiding rehashing them. Um relationships with causal inference. How do you establish

causality in these kinds of models? The the the nice thing is that whenever you construct a model like this like we're constructing uh if you can view a model kind of as a graph and then you can kind of start to look at the flow of it. Or whenever you look at hierarchical message uh message passing like we saw in chapter five we can actually see you know we can start to look at these models as like nodes and edges with a kind of directedness in the in the edges right so message passing matches uh messages from from one node to another or or from one variable to some other uh piece of the model whether it be you know some some kind of equation or or or conditional probability distribution or otherwise that connects them and relates them and allows for inference um or or or reaching kind of a the end or a kind of like delta or clamp. Um it is essentially I mean it's you know the sometimes modeling is as much an art as it is a science and I think many other people would be quite inclined to agree. Um and then a few more things you know we already know that this is heavily related to machine learning. We saw a lot of examples of other models in chapter 4 before we got to uh the what eventually we understood to be the POMDP model. It's like yeah if if you know you create a machine learning model or a linear regression model or something more simple but the idea of being able to take in values using some kind of transformation in order to uh you know elicit elicit some kind of output. Um that that's that's the broader theme. um robotics which is which has been very you know very interesting to see you know the sort of developments in robot I've spoken with a couple recent learners who had interest in uh excuse me I'll uh find a better way to this um but there is a I just I hope I can find this if I can't if there's anyone who's interested I'll I'll share again in future but um this is a few years ago so there's going to be a lot more by now um active inference robotics yeah so I mean there there there you know last year we did an entire symposium robotics was a theme um you know with the with the institute and then if you just kind of browse around this is 2019 right uh active inference for robot control and and there's just oh this again 2019 I so so it's just a because because I I think in the theme of chapter 10 of kind of the breadth of what active inference can touch upon or or cover or relate to or inform uh is this is just like looking at the breadth of you know these kinds of principles can be used like in real world settings uh for those who are interested in like you know business or enterprise or production or something like that I mean you can you know use them to create uh actual, you know, effective agents who who operate within the world, right? Whether whether it be some kind of, you know, real- time uh online learning, you know, model that's act, you know, repeatedly taking in data. It could be, you know, robotics. We know that from uh last year there was an active inference um like a a drone

uh that was built uh in and showcased though sadly there's no uh there probably is a public video. somewhere. But um got the uh yeah, directions of travel and active inference. Um anyway, I'm I'm jumping around a little bit, but it's just to show that these can relate to a variety of, you know, fields and of course many of the the folks who started Fristen and the rest are quite familiar with a lot of these other fields. So it's just to say that you know active inference is more so start it's it's starting a conversation with other fields. It's it's continuing conversations with other fields and all the while it's kind of establishing its own normative framework which in you know in my opinion is is highly interpretable because we have a capacity for relating uh you know a variety of functionalities and and formula and equations in models to actual terms like habits or or planning or or you know what what is you know perception and and these other kinds of things. So, um, is there anyone who would like to touch on anything in particular or if this was Yeah, John, please go for it. Uh, yeah. So, I have some concerns about some of the wording in this chapter and I'd like some clarification. Um, let's see. I have to doing the control S thing. Uh, What? Uh oh. See? Okay. Sorry. I thought I had this queued up. Um to go through again. [Music] So, okay. So um one of the one of the things I have concerns about is it says that one benefit of active inference is that it provides a complete solution to the adaptive problems that a sentient organism have to solve. Um what do you mean by that? Because you know for affordable system can't be complete and consistent among among other things like normally if I hear someone say they have a complete you know framework of everything that has to do with thought it sounds a bit odd so I wonder what you mean by complete. Mhm. No, good. Uh, no, good pointing that out. And I am also rather suspicious whenever someone, you know, it almost sounds like a sales pitch or something to say that something's, you know, full a complete solution, you know, end all be all. So here I mean I've always interpreted that as earlier on in the textbook they they they give this explanation you know whether it's necessary or not they give a explanation of uh what they call the the neats versus scruffies neats always seek unification beyond the heterogeneity of brain and mind phenomena which usually corresponds to designing top- down normative models that start from first principles. Meanwhile, but just because something's just because something's from first principles doesn't mean it's complete, right? All math is from first principles and no math can be complete. Yeah. Yeah. It's so they're not they're not necessarily saying they're not necessarily making some broad on kind of ontological or epistemological statement. I think from my understanding of this uh it's kind of a weird thing to say. So, I wonder what they mean by that cuz No, it's a good it's a good Well, what it does do, what I do

think they mean by it or or what kind of value there is to take away from that sentence. So, I I do kind of implicitly agree that maybe it could have been rewarded to to make it clearer on what they're going for. But with I brought up the neats versus scruffies bit because in the the scruffy's ideas that there are a lot of you know uh it's it's kind of like imagine a lot of researchers who are sort of you know going against the grain and they're they're moving in their own kind of unique direction and they're going after you know what whatever it is that's driving them or they're working with a team or or you know have some kind of funding or other whatever it is that's kind of drawing them towards wanting to hyperfocus or specialize on a particular phenomena potentially to the detriment of looking at other phenomena, then what happens is you can potentially end up with someone who's so terribly focused on perception that they leave out action or they're so focused on, you know, planning that they leave out attention. And so I my take on this is that it's it's trying to say a complete solution in the sense that, you know, the book is focused on something like sentient organisms. you know what what is the task that they're trying to solve. Um active inference is a framework where it's saying you know they're not trying to model necessarily the entire brain. It's kind of focused on a sort of beliefsbased framework where agents are trying to, you know, solve a a problem or a task, whether it be, you know, as simple as, you know, the autonomic system who who is who is breathing and and regulating blood oxygen flow to to a mouse in a T-mas or an entire society trying to function within an you know, an economic system. Uh and and so with that it's like well now we we have an understand it provides a kind of understanding or or solution for understanding all of these different phenomena right is it ever going to be fully complete I mean I that's you know I well no mathematically impossible yeah yeah if you want to take that hardline assumption that this is about mathematics but I don't know if this sentence is purely about mathematics is all I mean to I think it's more it's more about describing a kind of perspective and that's the issue right with with theory is that we're dealing in words and in sentences and full you know par that things can get kind of meshy for lack of a a better word. It's it's about as ambiguous as I think this statement could be interpreted as if looked at from particular angles. So okay but I I have concerns about this other phrase too and it's sort of the same concern because they kind of go together. It says uh what does it say? It says evolution appears to have discovered increasingly sophisticated design structures for brains and bodies that made organisms able to deal with and shape rich ecological niches. Fine. But then it says modelers can reverse engineer this process and specify brains and bodies of the creature of interest in terms of generative models. Like really? You know how to build a cat? It's so with

that let's see let me just make sure I understand it in context complex pursu Yeah. Yeah. Yeah. So I mean I mean in a sense yes to answer your question in a sense yeah I mean in so in chapter seven we look at the teammates example right like that's like a really simple thing though it's not like you you couldn't you couldn't build a cat brain in a laboratory using that or as far as I can tell any of this right like the idea that you're reverse engineering design and that you can specify the designed for brains and bodies like that that implies that you could you could make one. Yeah. I mean it's not I mean it's not it's not contradicting like Yeah. Yeah. I it is kind of saying that in a sense you're still ultimately modeling it, right? But you're so you do have to make decisions. But it kind of goes back to what I was saying earlier. It says you can specify the designs for brains and bodies. that that implies that you can design them because if you have a design, you can design it, right? Are are you are you trying to point out the the the separation between actually modeling, you know, a a a cat's brain, for example, versus the reality? And are you making an assumption that we'd never be able to do that? And therefore, well, no, it it doesn't matter if we'd ever be able to do it. The problem is, uh, we can't do it now. And so saying modelers present tense can reverse engineer this process and specify present tense the designs for brains and bodies the creatures of interest. That's it's a weird thing to say. That's like Victor Frankenstein almost like I I I don't I sorry I'm I'm not to be honest I'm not really sure how to address this much further because so in chap just to stick to the textbook chapter 7 we you know they make a mouse right they model a mouse they model it in a T-maze none of it's real so the formal technical term for that would be incilico and so you know that that is what they do and and if you. It's not the design of the brain in the body. It's just It's just a very It's just a very simple model of one aspect of something that a mouse can do. But it it its brain isn't its brain isn't I believe in that model, which is very simple there. The brain of the mouse and the body of the mouse aren't even even really in the model at all. It's just um they they they are and they're not. I I agree that in a literal sense, no, they're not. uh in this we're trying to sorry John and what we're trying to do with with active inference is that we're just trying to model the you know it is expressly what the the textbook states it's just trying to model the the problem that the brain is trying to solve in that instance. If you think that the problem that the organism is facing is much bigger, such as, you know, oh, it's not a mouse in a T- maze, it's a it's a mouse, you know, in in the woods or something, you know, and to where to where you have now you have to create like a broader environment. Maybe you create other agents who are in there as well. Maybe there's some kind of, you know, uh, world design where they they have movement that goes beyond just moving left or right or moving

to a queue. instead it's a much more like a broad like grid discrete space or maybe it's continuous space or or otherwise but I mean that that's a temp you know that's in a sense we are trying to anyone who's modeling something is trying to recreate its dynamics to better understand how it functions and so I just kind of quickly googled like cancer modeling it's like this the majority of research that's being done on cancer is involving computational modeling of cancer so Well, I do I think I do I kind of sympathize and understand maybe the point that you're making like you know but it's not to say that there's some kind of hubris of saying we're going to model the brain perfectly. We're going to reverse engineer and we're you know just assume that we can just magically do that. But it is making the point that like this is this is what most most people are doing in in a sense. I maybe it's not worded well. Maybe it's a little bit clever to say like, oh well, if if because they're relating it to other fields, right? So, we're looking at, you know, they relate it first to evolution. It's kind of like saying, well, evolution is what's kind of producing the the reality of of the organisms that we find in the world around us, but then from the other end, it's like a modeler is attempting to reverse engineer that. They're attempting to kind of create that or recreate that in in kind of like, you know, the opposite direction. No, I understand what modeling is, but I think my problem is really more with two words here. Specify and designs. Specify implies that it's specific. It's not just some general vague thing that that's like that that that simulates some aspect, but the specific design for the brain and the body. And then designs of course implies you can build one, but no, no one can build a cat. So, I'm just perplexed as to why uh it's it's like this. Okay. Yeah. I don't know. I think we might have to kind of agree to disagree on this one a little bit because I I don't fully I mean that that is, you know, that is what we do. I if if I sit down to kind of model uh I don't know I don't want to get too far away from the example we're talking about but if I want to model like I don't know the end of day stock price from financial data like I'm going to have to make some decisions on how to do that right I'm going to have to specify some things and I'm going to have to design some things in order to make that happen right so I think these are rather broad terms But you would specify the designs for plants which is easy. Specifying the design for a brain is is not easy. I don't think anyone can do it. And it it's, you know, not a not a thing I would say. Um I I don't there's there has been some interesting work. There's there's been some attempts to, you know, do this. I think there are plenty of people who would still agree with you, but I mean there there are folks who are who who at least have the aspiration and are even, you know, I know there's an aspiration. I sort of hear it myself, but but to to suggest that it's already been achieved is uh Yeah, if I can jump in. I Yeah. Yeah, please

do. Steve, I take John's part here. I'm very sympathetic um to what you're saying, John. Um I I also just assume there is so much not said or rather implied in the textbook that uh it I don't take it uh to a mathematical precision. The the words are an approximation of what is intended. Uh, and I I completely agree that this isn't a closed system. And if it were a closed system, uh, it could not be, um, completely described without, I mean, consistently completely described. Uh, and I really think what you're bringing up is worth bearing in mind always that um, a model is not the system itself. It's a model of the system. You know, designing saying that the brain can be reverse engineered uh is in I I just assume that means uh a an approximation of the architecture can be created uh based on this framework. I I I think too many people do have the assumption that yeah the brain can be created uh can be re represented and I just don't think that's possible. Uh the complexity is beyond the ability at the at this point in our technology. But it's I think intending to mean an approximation always. That's that's the my take on it. Yeah. Well, yeah. So, probably Okay. Oh, sorry. I was I I'm just seconding that. I mean, that that is essentially the what's going on with the modeling, whether it be an active inference or otherwise. But it but it is it is made explicit. I mean, I I agree with you both. the complexity of actually modeling the brain and not only that but what would be you know another qu a whole another question that comes up is what would be the the purpose of that I mean we we can maybe intuit it like oh it would be interesting or we'd find some kind of realization but it's I mean we start to get a little close to sci-fi right it's like well once you create the brain it does it's not really a brain of any particular person it's its own thing now and it's going to go off on its own trajectory so now is it even reflected is it going to be useful for simul ating or studying anything if you if it's if that makes any kind of sense. Well, you bring up a good point why it's really uh useless to try to create AGI uh in the way we think of it. I mean to create uh intell artificial intelligence that thinks just like a human you would have a human intelligence. So wait, we have that already. What's the point of that? Right. Right. Exactly. It's like, you know, it's it's so it would be superhuman immediately. Then it wouldn't be what we think of as intelligence. It would be something else. That would be wonderful, but it would be something else. I want to talk about this thing of modeling and approximation because like sort of everything in science is a model, but there are different kinds of things in science. There are models that are precise enough to build stuff with and there are models that aren't really precise enough to build like real world things. I'm not talking necessarily about computer programs. Um there there are models that aren't really precise enough to build real world things. And by saying that you can specify the design, you're you're suggesting that you

have enough precision to build real world things. But I I haven't seen uh I I haven't even seen that claim or anything like it made anywhere else in this book. Uh like I don't know. It's odd. Yeah, it's it's a it has a different meaning than what I think you take it as or it has we have to assume it means something different otherwise it wouldn't make any sense just as you're pointing out. Yeah, there there's a I'm mostly asking for clarification. I want to know what what that what does that mean? Why would you write that? Yeah, it it it is it is a strong piece that they you know that was quite a choice I guess for lack of a better phrase. I mean that they they expressly say earlier in the textbook like they are not trying to to recreate or even emulate the entire brain piece by piece is to find a generative model that describes the problem the brain is trying to solve. And that might not be, you know, that might not be a particularly satisfying sort of um that might not be a very satisfying take for those who are interested in doing something like trying to recreate the entire, you know, behavior of the of the I'm all for it. The whole thing. Sorry. I'm all for it if you can do it. just saying that you can do it when you can't is yeah they I think it's a it's a mistake for them. I think well I don't think that sentence should be read so literally. Um, but that that is that's kind of my final word from myself on my opinion on that because I think we're I'd just be belaboring it if I tried to in describing it more because the whole I mean if whenever I see that phrase for example complete solution uh you know if I think about in terms of mathematics it looks um ignorant or or or or just incorrect like why would you say that but if I look at at it from the perspective of kind of the neat scruffies bit, right? Where where there I mean you do have a broad variety of different departments doing very different things who have very strong debates with one another. Um if if there were a way to model the brain, you know, piece by piece as we were talking about earlier, I I imagine it would at minimum take many more of these departments coming together and agreeing upon a few things in order to even get going with something like that. Not to mention the kind of funding that might be necessary to realize something like that. Um, as long as fields stay broadly disparate, including how they use words, their ontologies, you know, I I think debates are fine, but but I I do appreciate in a sense the attempt to make uh kind of a broader framework. Is the word complete well used or or even relevant here? Probably not. But so the other thing is they do it twice. So on page um on page on page 196 they do it again. Um they say it that means active inference starts by providing a complete solution to the problems that organisms have to solve. So they say it twice to to the problems that organisms have. Okay. Sure. I Yeah. I I don't know. I I I think we're I don't know. Um I mean the question they said what they said is all right go ahead Stephen I mean just that

statement then immediately brings up the question in one's mind well in what sense are we talking complete solution and that's why mostly asking for clarification because I I assume it can't mean what it says. Yeah I I agree. Yeah, that's like author's author's framing. You know, the book you you write gets to have it how you think. And then but I'm also thinking like does a wing in an engine provide a complete solution to the problems faced by flight, right? Yes. But that doesn't mean that any given plane is going to fly or any given company is going to work. It's like it's not a guarantee. But it's like but yes between lift and thrust that between sensemaking and cognition and action it is like the scheme of a complete solution to the problems of the kind that they do face. Yes. Any given inquiry is going to be successful but but it it it is like okay the complete meal is the main dish and the side dish doesn't mean you're going to have a successful restaurant but it is a complete approach to cooking dinner if it has those two components. Yeah, it's like the idea of, you know, perception without action, right? Is so that was my attempt to kind of contextualize some of these things which, you know, there there's an entire sub field known as intellectual history, right? that that that people try to like do this kind of hermeneutical or exeetical approach to the study of literature including psychology or otherwise looking at the ontologies extracting them relating them to one another contradictions and and all of that and so it's kind of like active inference is kind of placing itself in a certain part of that trajectory and you know they they have their own take and they're claiming I maybe maybe they should use the phrase more complete I think that would get closer to maybe what they're actually going for um you I'd be happy with that. Then hyper desperate. Yeah. Yeah. Um but we're not the authors, so um yeah, so it goes. Yeah. Food for thought, but it's still I mean it's still a good question, right? Because it does open up onto these bigger questions and and then um and and also to Stephen's point earlier, right? Like at the end of the day, we're still making proxies. So if you if you quote unquote design or specify or design the brain of the the mouse and the T-mas it's like well no you're not really but in a sense you you are in this only in the sense of if active inference is making the claim that that the brain or does something like uh perception and action as we're we've described it and looked at state inference and policy inference and so on then it kind of like you kind of are because the mouse is carrying out the the hypothetical encilico mouse is carrying out those processes, right? We're just kind of running the code that represents that. Well, I thought it would but but just Andrew in the last um four minutes, could you look at read Isabella's question up in the chat and just give a first anyone can give a first thought on that and then we'll end in a few minutes. Yeah, absolutely. Uh no, and thanks for your question Isabella.

What is the relationship of active inference and autopoietic systems? It just occurred to me robots can perform active inference while not being autopoietic and there are living systems without central nervous systems. Like what does it mean about the relationship between active inference and life? So let's just um we do see this term in the textbook but just to like get a quick like autopoiesis self-making a system's ability to self-organize self-reproduce and self-maintain through its own internal processes. So one interpretation of that is you know to ensure their own reproduction and maintenance. Well, with robots, yeah, not so sure about the reproduction part. Maintenance potentially, right? Like like you could have a robot that maybe it doesn't necessarily reproduce, but it might have actions that are available to it that it could then use to or have available to it to take care of itself. But I mean that's the that's kind of what separates the the the claim in the textbook about what evolution is led towards such as beings such as us who who who who are you know able of not just reproducing but also full maintenance versus a robot that you know if it if it if it doesn't have arms to reach for the the power cord to plug itself in or something along those lines, right? I mean it's kind of a silly way of saying that but but in any case but that is a broader philosophical question as well. um you know what is the relationship between this and life? Um there are you know there are many people who have made claims about things like pansychism uh things like you know well if you if really something like sentience or something like you know active inference related processes uh are just kind of like mathematical formalisms that are playing out before our eyes thanks to you know the Hamiltonian pathist action uh or or other kind of physical processes then is it the case that many things that we didn't think of as sentient before in fact are uh is it not the case that like a rock rolling down the hill because of a variety of is it not doing some kind of state inference I that these are larger debates that one can engage in um but uh I I personally I don't really follow that line so much um but but there has been some really kind of philosophy of mind active inference Yeah, a reference in the chat that is related to that. The the uh handbook of abductive cognition uh kind of gets at this whole epistemological issue of what are what is cognition? Uh how do we know what we know? How why do we call something sentient and not other things? uh you know a self uh regulating system that is not even biological uh you know chemical processes. We don't have to imbue any sentience to that. It's just a natural process in and just in the same way that biological systems are natural processes but biological systems interact with the environment in adaptive ways in self-generate self-perpetuating adaptive ways. I I think it Isabella's question is uh relevant in that u it it uh active inference is a way of describing processes in any framework we want to place it

if it is self-perpetuating or not in a you know evolutionarily versus in this closed time frame you know in this single environment experiment closed time frame. We're still talking about the same natural processes. Uh I think um variational free energy uh is describing a system and we just decide what that system is that we're applying it to. Yeah. And it's a great a great connection with the abductive logic and the pragmatism and the open-endedness of evolution because it's like the success of generations intergenerationally or successive beliefs or hypotheses intragenerationally like behavior. They aren't just induction deduction elevator rides. They're moving laterally and generating new pieces and new compositions in the sort of open-ended reconfiguration and then the niche sweeps it off the table if it doesn't maintain autopoiesis in actuality. But that doesn't mean that every map you make of that territory is describing its autopoiesis. Like in the T-maze, it's just taken for granted we're talking about a behaviorally alive mouse. Yes, we're not studying the autopoiesis of that mouse. But that might be another inquiry. Yeah. Yeah. Is a classic kind of phrasing that's been used in active inference and elsewhere. But the idea of like, you know, the map is not the territory. And that relates back to, you know, what we were talking about earlier with uh with modeling. Anyway, we got kind of, you know, broad and it's, you know, in reference to that bit about overcoming, you know, old categories and 19th century kind of constructions. It's like the notion of sentience. Like, I'm not sure if going back to read William James or or even something from the mid 20th century actually is going to align well with whatever it is that we're discovering in modern science. Same thing for any notion of things like consciousness, right? I mean it for some consciousness is literally just like you know you're going to put someone under anesthesia you need to make sure they're unconscious you use you know an EEG you know headset or or otherwise to try and measure quant consciousness is like a quantitative signal and that's it right it's not even it's not a big broad philosophical category it's literally just a pragmatic practical concept that is useful there in medicine uh others will say you know dramatically different things. So, it's uh always uh always interesting conversation. Uh in any case, we're a few minutes over. So, thank you all so much for your for your question. Sorry, Isabella for only getting to the question at the end, but I hope that that uh kind of subcon conversation was helpful in in some kind of way. Thanks. Also, I'm just looking at this book. \$699 for the ebook. Let's get it. It's worth it. And it's hard to get copies. I have to do library loan uh you know scan chapter at a time. Wow. Cool. All right. Thank you all. Bye. Thanks everyone.